

QA/4HANA Studio

PRODUCT DOCUMENTATION · CURRENT VERIFIED BUILD

User Training Work Instruction

Detailed screen-by-screen operating instruction for workspace owners, functional consultants, test authors and execution operators.

DOCUMENT Q4H-WI-USER-2.0	PRODUCT BASELINE QA/4HANA Studio 2.1.0 development line; v2.0.0 remains frozen
PREPARED 30 July 2026	CLASSIFICATION Internal product and training documentation

Screens containing configured target or account information have been anonymized. Passwords and credential values are never included.

BEFORE YOU BEGIN

Audience, prerequisites and safety

Audience

- Workspace owner or administrator
- Functional consultant / Test author
- Automation engineer
- Execution operator
- Auditor or evidence reviewer

Prerequisites

- Access to the Studio URL
- A preserved owner account
- A configured non-production SAP target
- Authorized test data and business references
- Approved permission for transactional runs

Safety rule. A failed transactional Test stops at the failing stage and retains evidence. Do not automatically reverse or delete the business transaction; doing so would remove the compliance trail. Any remediation or rerun requires a new controlled execution record.

Navigation conventions

Control	Meaning
Coral primary button	Creates, saves, advances or starts the primary action.
Previous / Next	Moves through the workspace sequence; it does not automatically save unsaved work.
Target badge	Shows configured application, target class and verification state; select it to review target configuration.
Profile name	Opens Settings, Help and Sign out.
Draft / Published	Draft permits editing; Published indicates validation has succeeded for governed use.

SCREEN-BY-SCREEN INSTRUCTION

1 Automation Overview

Use the landing canvas to understand workspace inventory, actual execution outcomes and modeled automation value, then resume or start work.

QA/4HANA Studio by aisk

Workspace / Automation Overview

SAP - configured-sap-target.example - Non-production - Verified

YOUR AUTOMATION WORKSPACE

Good morning

Continue recent work, review executions, or start a new test.

+ Create test ▷ New execution

11 Tests 5 Business Processes 53 Controls 178 Evidence records

MEASURED OUTCOMES AND MODELED IMPACT

Execution impact

Run totals and duration are exact. Manual effort and cost are transparent scenario estimates. Recent runs →

Total executions ACTUAL 89 Recorded in the immutable run ledger	Execution outcomes ACTUAL 58 Passed 31 Failed 	Automation runtime ACTUAL 3.9 h Sum of recorded execution durations
Plausible manual effort MODELED 26.8 h Scenario range 18.0–44.8 h	Potential time saved MODELED 22.9 h Manual estimate minus automation runtime	Potential cost saved MODELED -US\$283 US\$1,340 manual equivalent – US\$1,623 automation TCO – Investment not yet recovered

► Calculation assumptions and model

Pick up where you left off View all →
 Saved test cases available in this workspace

TEST CASE
Cleanup Abandoned Drafts

Automation Overview · anonymized current-build capture

Procedure

- 1 Confirm the target badge.**
 Check the application, non-production classification and Verified state.
 Expected: The badge reflects the intended SAP test target.
- 2 Review inventory totals.**
 Read Tests, Business Processes, controls and evidence record counts.
 Expected: Counts reflect current persisted workspace data.
- 3 Review execution impact.**
 Separate ACTUAL metrics from MODELED manual effort, time and cost scenarios.
 Expected: Recorded and modeled values are not confused.
- 4 Inspect calculation assumptions.**
 Expand the assumptions section when validating the modeled business case.

5

Choose the next task.

Select Create test, New execution, Recent runs or a recent saved artifact.

Completion check. You can explain the current workspace volume, recent outcome and next action without navigating date folders.

Watch for:

- A negative potential cost saved value means the modeled automation investment has not yet been recovered.
- Modeled metrics are scenarios, not audited financial savings.

SCREEN-BY-SCREEN INSTRUCTION

2 Control Object Repository

Use the repository to scan a configured SAP screen, capture reusable controls and maintain the selector inventory.

The screenshot shows the QA/4HANA Studio interface. On the left is a sidebar with navigation items: Automation Overview, Control Object Repository (highlighted), Compose, Test Data, Processes & Packs, Execution Center, and Audit and Evidence. The main area is titled 'Workspace / Control Object Repository'. At the top right, there's a status bar showing 'SAP · configured-sap-target.example · Non-production · Verified' and navigation buttons 'Previous' and 'Next'. Below this, the 'Scan a live screen' section contains a URL input field with 'https://my426318.s4hana.cloud.sap/', an 'App ID, e.g. procurement' input field, and an 'Open scan session' button. A note states: 'The initial URL comes from Settings → Test-system connections → SAP. You can append or replace the Fiori route for this scan.' Below the note is a text box showing 'Target: Non-production · verified 29/07/2026, 21:34:27'. The 'Browse objects' section has a description: 'Every App ID with at least one saved object, grouped by domain — check here before capturing a screen again under a new name.' and a dropdown menu labeled '— select a domain (3) —'. At the bottom left, a 'Workspace owner' section shows 'Local single-user workspace'. A footer bar at the bottom reads 'Control Object Repository · anonymized current-build capture'.

Procedure

- 1 Verify the scan URL.**
The initial URL is populated from Settings → SAP. Add or replace only the permitted Fiori route.
- 2 Enter or confirm the App ID.**
Use the business application identifier that owns the screen.
- 3 Open the scan session.**
The Studio opens a controlled browser session and authenticates with the stored SAP connection.
Expected: The intended non-production SAP screen opens.
- 4 Capture controls.**
Use bulk scan or continuous Ctrl/Cmd interactive picking. Review duplicate warnings before saving.

5

Curate captured objects.

Confirm business name, selector/type, domain, App ID and ordering.

6

Browse saved objects.

Filter by domain and App ID, open the stable object route, and use Highlight to recheck a live control.

Completion check. Each saved object has an explicit App ID, reusable identity and a stable repository route.

Watch for:

- Never enter SAP passwords on the scan screen; credentials come from Settings.
- A “not found” highlight requires selector review—it is not proof that the business application is unavailable.

SCREEN-BY-SCREEN INSTRUCTION

3 Compose — Test Library

Use the Test Library to find existing Tests by business meaning and avoid duplicate automation assets.

Workspace / Compose

SAP · configured-sap-target.example · Non-production · Verified

← Previous Next →

Reusable automation assets

Test Library

Find a Test by business meaning, application, process area or readiness.

+ New Test

SEARCH: Name or file

FILTER BY PROCESS AREA: All process areas

FILTER BY APPLICATION: All applications

FILTER BY STATUS: All statuses

11 of 11 Tests

TEST	PROCESS AREA	APPLICATION	STEPS	STATUS	
Cleanup Abandoned Purchase Order Drafts cleanup-abandoned-drafts.json	Generic	SAP	2	LEGACY READY	Open Test
Create Billing Document create-billing.json	Purchasing	SAP	12	LEGACY READY	Open Test
Create Outbound Delivery and Post Goods Issue create-delivery.json	Sales	SAP	19	LEGACY READY	Open Test
CREATE PURCHASE ORDER - MISSING LINE ITEM IS BLOCKED create-po-missing-item-negative.json	Untagged	SAP	6	LEGACY READY	Open Test
Create Purchase Order - Happy Path create-po.json	Purchasing	SAP	7	LEGACY READY	Open Test
Create Sales Order - Happy Path create-so.json	Sales	SAP	16	LEGACY READY	Open Test
Open Manage Purchase Orders - Read Only open-manage-purchase-orders.json	Untagged	SAP	1	LEGACY READY	Open Test
Post Goods Receipt for Purchase Order post-goods-receipt.json	Purchasing	SAP	7	LEGACY READY	Open Test
Post Supplier Invoice (3-Way Match)	Purchasing	SAP	12	LEGACY READY	Open Test

Workspace owner
Local single-user workspace

Compose — Test Library · anonymized current-build capture

Procedure

- 1 Search before creating.**
Search by Test name or file and filter by process area, application and readiness.
- 2 Open an existing Test.**
Select Open Test on the matching row.
Expected: The browser moves to a stable `/compose/tests/...` route.
- 3 Create a new Test.**
Select New Test and enter business name, generated or edited file name, application and process area.
- 4 Choose a starting point.**
Use Blank for new behavior or Template to copy an existing Test without modifying the source.

5

Create the Test.

Resolve any duplicate-file warning before continuing.

Completion check. The Test opens on a refresh-safe route and is visible in the library with its readiness status.

Watch for:

- File names are technical stable identifiers; use the business name for readable purpose.
- Template creation copies steps and contract—it does not establish execution lineage.

SCREEN-BY-SCREEN INSTRUCTION

4 Compose — Test editor

Define application metadata, typed inputs/outputs and executable steps, then validate and publish the reusable Test.

Workspace / Compose

← Back to Test Library

▶ How Objects, Modules, and App ID fit together

Test lifecycle
Legacy ready
Publishing requires a valid contract, parameters and objects. DRAFT

TEST CASE NAME: Create Purchase Order - Happy Path

APPLICATION: SAP

DOMAIN: Purchasing (Generic, Purchasing, Sales)

No declared Test contract
Legacy inference keeps this Test executable, but publishing requires reviewed typed inputs and outputs. Use inferred contract

Steps (7)

#	MODULE	APP ID	PARAMS				
1	CreateAutomationRunReference		prefix=\${automationReferencePrefix} . owner=\${automationOwner} . maxLength=16	↑	↓	Edit	Remove
2	Login		url=\${url} . username=\${username} . password=\${password}	↑	↓	Edit	Remove
3	NavigateToApp		url=\${urlBase}/ui#PurchaseOrder-manage	↑	↓	Edit	Remove
4	ClickButton	CREATEPURCHASEORDER	control=CreateButton	↑	↓	Edit	Remove
5	EnterHeaderField	CREATEPURCHASEORDER	field=SupplierField . value=\${supplier}	↑	↓	Edit	Remove

Workspace owner: Local single-user workspace

Compose — Test editor · anonymized current-build capture

Procedure

- Review lifecycle and metadata.**
Confirm Test case name, application and domain.
- Author the Test contract.**
Declare typed inputs and outputs; classify sensitivity where business or secret values are involved.
- Use inferred contract carefully.**
Legacy Tests can infer placeholders, but publishing requires a reviewed contract.
- Add or edit steps.**
Select a module, supply required parameters, choose reusable objects and set literal, dataset, system-context or prior-output value sources.

5 Reorder with accessible controls.

Use the provided up/down keyboard-operable actions; do not depend on drag alone.

6 Save the Test.

Save draft changes before leaving the route.

7 Publish the Test.

Resolve unresolved inputs, missing objects/parameters or output collisions until validation succeeds.

Expected: Lifecycle changes to Published or the editor shows an actionable blocking finding.

Completion check. The Test round-trips to executable ModuleCall JSON and has a reviewed contract suitable for Process/Pack binding.

Watch for:

- Do not place passwords or authorization tokens in literals, Test contracts or datasets.
- Changing an output name can break downstream Business Process hand-offs.

SCREEN-BY-SCREEN INSTRUCTION

5 Test Data

Create reusable flat, nested or relational data while preserving transaction ownership and previewing effective structures.

The screenshot shows the QA/4HANA Studio interface with the 'Test Data' workspace selected. The left sidebar contains navigation options: Automation Overview, Control Object Repository, Compose, Test Data (highlighted), Processes & Packs, Execution Center, and Audit and Evidence. The main workspace area is titled 'Workspace / Test Data' and shows a status bar with 'SAP · configured-sap-target.example · Non-production · Verified'. Below this, there are two tabs: 'OPEN DATASET' (selected) and 'OR CREATE NEW'. The 'OPEN DATASET' tab shows a dropdown menu with '— select —'. The 'OR CREATE NEW' tab shows a dropdown menu with 'Flat CSV' and a 'Create' button. Below these tabs, there is a section titled 'Relational CSV builder' with the subtitle 'One header · many owned children'. It includes a description: 'Join two CSV files without flattening line items. Validation blocks duplicate header keys, missing keys, orphan children, and collection-name collisions.' The interface has several input fields: 'OPEN RELATIONSHIP' (dropdown with '— New relationship —'), 'RELATIONSHIP NAME' (text input with 'sales-orders-with-items'), 'HEADER CSV' (dropdown with 'Select header file'), 'HEADER KEY' (text input with 'scenarioKey'), 'CHILD CSV' (dropdown with 'Select child file'), 'CHILD FOREIGN KEY' (text input with 'scenarioKey'), and 'CHILD COLLECTION NAME' (text input with 'items'). There are also 'Validate relationship' and 'Save relationship' buttons. At the bottom, a status bar indicates 'Test Data · anonymized current-build capture'.

Procedure

- 1 **Choose the model.**
Select Flat CSV, nested JSON or relational CSV based on the transaction structure.
- 2 **For flat CSV, define rows and columns.**
Preserve headers and use proper CSV quoting for commas, quotes and line breaks.
- 3 **For nested JSON, keep children under their owning header.**
One sales order with many line items is one transaction; multiple orders are separate transactions.
- 4 **For relational CSV, configure the relationship.**
Select header file/key, child file/foreign key and child collection name.
- 5 **Validate and preview.**
Resolve duplicate headers, missing keys, orphan children or collection-name collisions.

6

Save and reopen.

Confirm the stable dataset route and preview are unchanged after reload.

Completion check. The saved model preserves owned children and can be selected during execution preflight.

Watch for:

- Never flatten multiple headers and line items into an ambiguous Cartesian set.
- Treat sensitivity metadata and dependency-safe rename/removal as controlled governance work until the remaining backlog is completed.

SCREEN-BY-SCREEN INSTRUCTION

6

Processes & Packs — workspace entry

Choose whether you are composing a sequential Business Process or an independent Regression Pack.

The screenshot shows the QA/4HANA Studio interface. The left sidebar contains a navigation menu with the following items: Automation Overview, Control Object Repository, Compose, Test Data, Processes & Packs (highlighted), Execution Center, and Audit and Evidence. The main workspace area is titled 'Workspace / Processes & Packs'. At the top right of the workspace, there is a status bar showing 'SAP · configured-sap-target.example · Non-production · Verified' and navigation buttons for 'Previous' and 'Next'. Below the title bar, there are two tabs: 'Business Processes' and 'Regression Packs'. A descriptive text states: 'A Business Process sequences Tests and carries outputs forward between stages.' Below this, there are two sections: 'OPEN BUSINESS PROCESS' with a dropdown menu showing '— select —', and 'OR CREATE NEW' with a text input field containing 'po-gr-invoice' and a 'Create' button. At the bottom left of the workspace, there is a 'Workspace owner' section showing a user icon and the text 'Local single-user workspace'. A footer bar at the bottom of the workspace displays the text 'Processes & Packs — Business Process entry · anonymized current-build capture'.

Procedure

1

Select Business Processes for sequential work.

Use this when later Tests require outputs from earlier Tests.

2

Select Regression Packs for independent work.

Use this to group Tests or Processes that do not hand data to one another.

3

Open an existing artifact.

Use the file picker; the selected artifact receives a stable route.

4

Create a new artifact.

Enter a technical file name, create it, then supply its business name and lifecycle metadata.

Completion check. You have selected the correct topology before adding Tests or data bindings.

Watch for:

- Do not use a Pack to model sequential hand-offs.
- Do not use a Business Process merely as a folder for unrelated Tests.

SCREEN-BY-SCREEN INSTRUCTION

7 Business Process editor

Order reusable Tests into a governed scenario and map data or prior-stage outputs into each stage.

QA/4HANA Studio by aisk

Workspace / Processes & Packs

SAP · configured-sap-target.example · Non-production · Verified

← Previous Next →

Business Processes Regression Packs

A Business Process sequences Tests and carries outputs forward between stages.

OPEN BUSINESS PROCESS

po-gr-invoice.json

OR CREATE NEW

po-gr-invoice Create

BUSINESS PROCESS NAME (SCENARIO TITLE)

Create PO - GR - Invoice

DOMAIN

Purchasing Save

Generic Purchasing Sales

LIFECYCLE

Draft

Test cases (3)

AVAILABLE TEST CASES

Filter...

cleanup-abandoned-drafts.json + Add

create-billing.json + Add

create-delivery.json + Add

create-po-missing-item-negative.json + Add

create-po.json + Add

create-so.json + Add

BUSINESS PROCESS ORDER

1 create-po.json ↑ ↓ ×

2 post-goods-receipt.json ↑ ↓ ×

3 post-supplier-invoice.json ↑ ↓ ×

Workspace owner Local single-user workspace

APP ID DATA FILE

Processes & Packs — Business Process editor · anonymized current-build capture

Procedure

- 1 Set business name, domain and lifecycle.**
Keep the Process in Draft while changing topology.
- 2 Add Tests in business order.**
Select Tests from the available list and use up/down controls to establish the stage sequence.
- 3 Confirm App ID and data binding.**
Select the application and transaction data required by the Process.
- 4 Map stage inputs.**
Bind each required input to process data, a literal, system context or a prior-stage output.
- 5 Review hand-off validation.**
Resolve forward/cyclic references, unknown outputs, duplicate outputs and type mismatches.

6

Save, then publish.

Publish only when all required contracts and policies are valid.

Completion check. Every stage has an explicit ordered identity and each required input resolves from an allowed source.

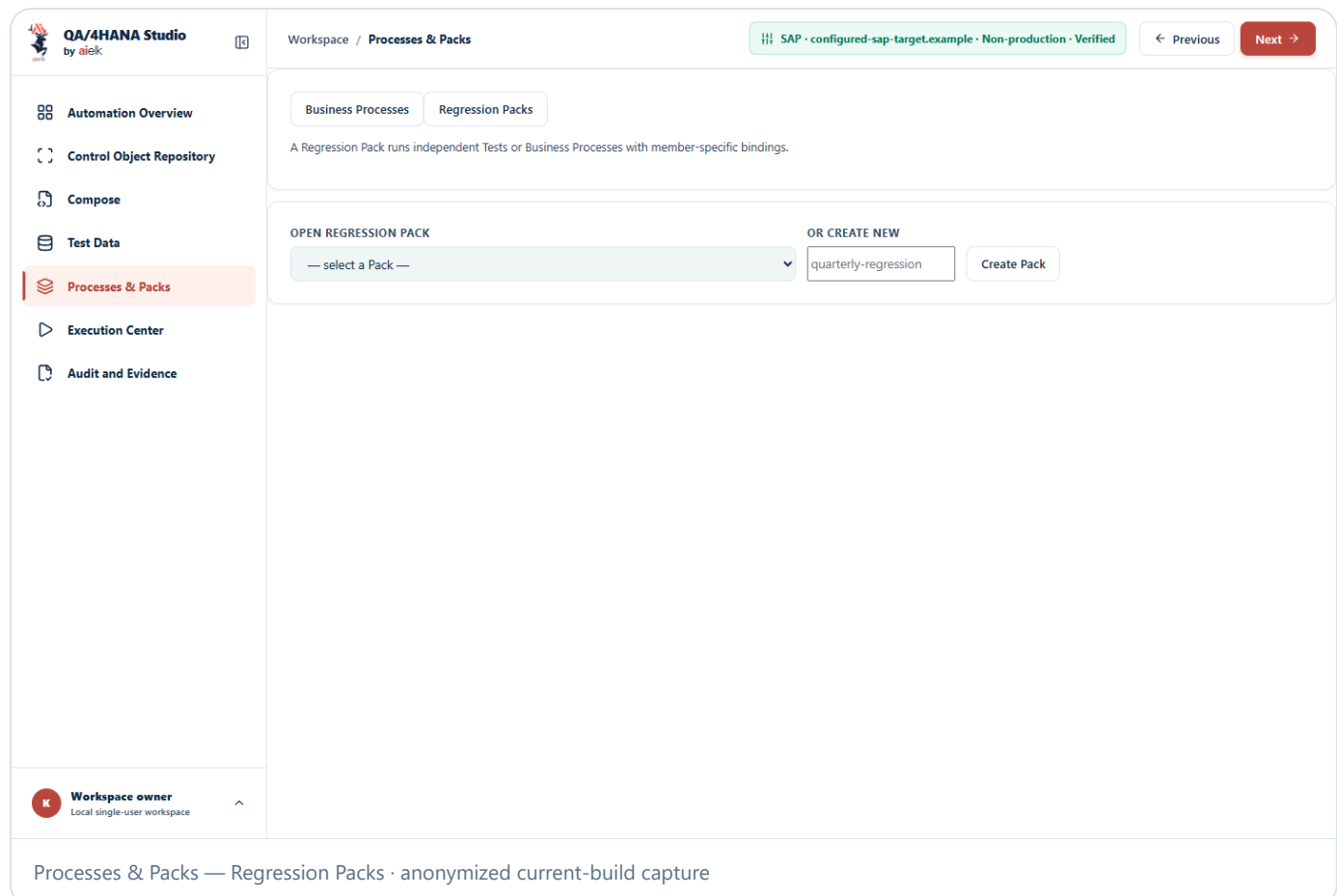
Watch for:

- A failed stage stops later stages for that transaction.
- Outputs are namespaced by stage; avoid reusing ambiguous names.

SCREEN-BY-SCREEN INSTRUCTION

8 Regression Pack editor

Create an independently executable collection of Tests and/or Business Processes with member-specific policies.



Procedure

- 1 Create or open a Pack.**
Give it a business name, version and Draft lifecycle.
- 2 Add independent members.**
Add Test or Business Process members; each member receives a stable ID.
- 3 Set member bindings.**
Configure member data, application and allowed session/failure behavior.
- 4 Reorder for reporting intent.**
Order does not create cross-member data hand-offs.
- 5 Validate references.**
Resolve missing artifacts, data files, duplicate IDs and incompatible policies.

6**Publish the Pack.**

Only published Saved Packs are eligible for governed execution.

Completion check. The Pack contains independent, valid members and can be selected as a Saved Pack in Execution Center.

Watch for:

- Test members cannot use “reuse within process” session behavior.
- Published Packs may not reference Draft Business Processes.

SCREEN-BY-SCREEN INSTRUCTION

9 Execution Center

Prepare and authorize a controlled run without opening SAP until scope, data, target and policies pass preflight.

The screenshot displays the QA/4HANA Studio interface, specifically the Execution Center. The left sidebar contains navigation options: Automation Overview, Control Object Repository, Compose, Test Data, Processes & Packs, Execution Center (highlighted), and Audit and Evidence. The main area is titled 'NEW EXECUTION' and 'Prepare a controlled SAP run'. It includes a progress bar with four steps: 1. Scope, 2. Data & policies, 3. Preflight, and 4. Review. The 'Execution type' section shows 'Single Test' selected. Below this, 'AVAILABLE TESTS' lists several JSON files with '+ Add' buttons. The 'SELECTED TEST' section is currently empty. The 'APP ID' field shows 'createPurchaseOrder'. The 'DATA MODEL' dropdown is set to 'Single CSV or nested JSON'. The 'TRANSACTION DATA FILE' dropdown is set to 'none'. A 'HEADLESS' checkbox is present. At the bottom, a status bar indicates 'Execution Center — new controlled run · anonymized current-build capture'.

Procedure

- 1 Select execution type.**
 Choose Single Test, Business Process, Pack · Tests, Pack · Processes or Saved Pack.
- 2 Define Scope.**
 Select the exact Test/Process/Pack and confirm App ID.
- 3 Set Data & policies.**
 Choose the transaction data model/file, filters, limits, headed/headless behavior and permitted failure/session policies.
- 4 Run Preflight.**
 Review target, credentials, artifacts, versions, contracts, effective data, mappings, calculated work and safety findings.

5

Correct blocking findings.

Use exact artifact links where available, save corrections and rerun preflight.

6

Review immutable scope.

Confirm members, iterations, stages, steps, child work, data snapshot and plan hash.

7

Acknowledge warnings and start.

Begin only when the target, owner reference and transactional safeguards are approved.

8

Monitor progress.

Use the stable run route; reconnecting rebuilds persisted top-level progress.

Completion check. The run has a unique immutable ID, exact input snapshot, controlled target and visible progress/evidence state.

Watch for:

- Do not start transactional Tests against an unclassified or production target.
- On failure, stop downstream work and retain evidence; do not auto-reverse.
- A rerun creates new lineage—it never overwrites the original run.

SCREEN-BY-SCREEN INSTRUCTION

10

Audit & Evidence

Find an immutable run without browsing date folders and open the single canonical evidence document.

QA/4HANA Studio by a1sk

Workspace / Audit and Evidence

SAP · configured-sap-target.example · Non-production · Verified

← Previous Next →

IMMUTABLE EXECUTION RECORDS

Audit and Evidence

Find a run by process, application, outcome, date, run ID, or executor—without navigating date folders.

Refresh

🕒 89 Total runs ⌚ 58 Passed 🚫 31 Failed

🛡️ Canonical evidence is owner-protected and redaction is enforced
Retention: until the workspace owner deletes it · credentials excluded · execution logs filtered

🔍 Search All outcomes All modes All dates

89 runs Newest first

CREATE PURCHASE ORDER - MISSING LINE ITEM IS BLOCKED PASSED

Jul 30, 2026, 6:40 AM GMT+5:30 1m 17s chain createPurchaseOrder Workspace owner

CREATE PURCHASE ORDER - MISSING LINE ITEM IS BLOCKED

299669f8-75b6-471d-9c56-ed8375811fc5

CREATE PURCHASE ORDER - MISSING LINE ITEM IS BLOCKED FAILED

Jul 29, 2026, 9:35 PM GMT+5:30 1m 28s chain createPurchaseOrder Workspace owner

CREATE PURCHASE ORDER - MISSING LINE ITEM IS BLOCKED

eb868bc6-673b-4e65-8919-6ccf66b6a645

Create Sales Order - Happy Path PASSED

+2 linked test cases

Jul 29, 2026, 4:29 PM GMT+5:30 6m 59s chain createSalesOrder Workspace owner

Create Sales Order - Happy Path Create Outbound Delivery and Post Goods Issue Create Billing Document

4b4f15d7-32df-4348-b678-23fe5243ff7a

View record Evidence PDF

Workspace owner
Local single-user workspace

Audit & Evidence · anonymized current-build capture

Procedure

1

Search by business identity.

Search process/Test name, App ID, run ID or executor.

2

Apply outcome, mode and date filters.

Use the current client filters to narrow the newest-first list.

3

Open the run detail.

Confirm stable route, outcome, duration, executor, target context and source artifacts.

4

Open Evidence PDF.

Use the canonical link. Execution Center references this same document.

5

Review lineage and retention.

Confirm parent/rerun identity, redaction posture and retained failure evidence.

Completion check. The reviewer can identify the run and open one canonical branded evidence PDF with no duplicate version.

Watch for:

- Server-side pagination and complete lineage traversal remain future work.
- Evidence can contain business data; follow the displayed retention and access policy.

ACCOUNT MENU INSTRUCTION

11 Settings, Help and account actions

Select the workspace-owner name at the lower-left to open Settings, Help or Sign out. The Settings screenshot is intentionally omitted because it can display configured connection identity.

1 Open the profile menu.

Select the owner name/avatar.

2 Open Settings.

Choose appearance and review connection panels for SAP, Salesforce, Oracle and ServiceNow.

3 Configure SAP.

Enter the non-production test-system URL, username and password, then select Save SAP connection.

Expected: the URL/username and non-secret credential status persist; the password is not returned to the browser.

4 Verify the target.

Classify the target and perform the non-transactional verification required by the workspace context.

5 Use Help.

Open contextual help from the profile menu.

6 Sign out.

Use Sign out when leaving a shared device. Workspace artifacts remain associated with the preserved owner identity.

Credential rule. Never copy passwords into screenshots, evidence, Tests, datasets, logs or support messages. The password field is write-only from the browser's perspective.

RECOMMENDED PRACTICAL ASSESSMENT

12 End-to-end supervised training exercise

- 1 Verify a non-production SAP target.**
Record the authorized owner reference.
- 2 Capture one reusable control.**
Confirm App ID and duplicate check.
- 3 Create one Draft Test.**
Add a typed input, one reusable object and a safe read-only step.
- 4 Create a small dataset.**
Preview the effective record.
- 5 Publish the Test.**
Resolve all publish findings.
- 6 Create a two-stage Business Process.**
Map one prior-stage output to a later input.
- 7 Create a Regression Pack.**
Add the Test and Process as independent members.
- 8 Preflight a headed non-transactional execution.**
Review snapshot, plan hash and calculated work.
- 9 Run and monitor.**
Reconnect once to demonstrate stable run recovery.
- 10 Open canonical evidence.**
Find the run from Audit & Evidence and explain its lineage and retention.

Training pass condition. The learner completes the flow without exposing credentials, bypassing preflight, confusing Process hand-offs with Pack independence, or altering the original run during a rerun.